Chapter 15. Multicollinearity and Biased Estimation

Multicollinearity, VIF, ridge regression, principal component regression

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$\hat\beta = (X'X)^{-1}X'\mathbf{y}$ is BLUE with covariance $\sigma^2(X'X)^{-1}$, huge when $X'X$ is nearly singular; a slightly biased, much less variable estimator can then be far more accurate.

We standardize: y centred, regressors centred and scaled ($\sum_i x_{ij} = 0$, $\sum_i x_{ij}^2 = 1$), so X is $n \times p$, $X'X$ is the regressor correlation matrix and $X'\mathbf{y}$ is proportional to their correlations with y. Original-scale coefficients: $\tilde\beta_j = \hat\beta_j\, s_y/s_j$, $\tilde\beta_0 = \bar y - \sum_j \tilde\beta_j \bar x_j$.

15.1 Multicollinearity

Multicollinearity means that the regressors are (nearly) linearly dependent. Exact dependence makes X rank-deficient and $\hat\beta$ undefined; near dependence makes $(X'X)^{-1}$ huge and $\hat\beta$ unstable.

Example 15.1 (Two regressors)

With two standardized regressors with correlation r_{12},

$$X'X = \begin{pmatrix} 1 & r_{12} \\ r_{12} & 1 \end{pmatrix}, \qquad \operatorname{Var}\hat\beta = \frac{\sigma^2}{1-r_{12}^2}\begin{pmatrix} 1 & -r_{12} \\ -r_{12} & 1 \end{pmatrix},$$

so $\operatorname{Var}\hat\beta_j = \sigma^2/(1-r_{12}^2) \to \infty$ as $|r_{12}| \to 1$, and $\operatorname{Corr}(\hat\beta_1, \hat\beta_2) = -r_{12}$. With $r_{12} = 0.99$ the variances are 50 times those under orthogonal regressors, and confidence intervals $\hat\beta_j \pm t_{\alpha/2}(n-3)\sqrt{MSE/(1-r_{12}^2)}$ are 7 times as wide.

Eigenvalues. If $\lambda_1, \dots, \lambda_p$ are the eigenvalues of $X'X$,

$$\sum_j \operatorname{Var}\hat\beta_j = \sigma^2 \operatorname{tr}(X'X)^{-1} = \sigma^2 \sum_j \frac{1}{\lambda_j},$$

so a single small eigenvalue blows up the total variance. The eigenvector of a small eigenvalue shows which combination of regressors is nearly constant.

Theorem 15.1 (Variance inflation factor)

Let R_j^2 be the coefficient of determination of the regression of x_j on the other regressors. Then c_{jj}, the j-th diagonal entry of $(X'X)^{-1}$, equals $(1-R_j^2)^{-1}$, so

$$\operatorname{Var}\hat\beta_j = \frac{\sigma^2}{1-R_j^2}.$$

Proof. Write $X = (\mathbf{x}_j, X_{-j})$. By the partitioned-inverse formula,

$$c_{jj} = \{\mathbf{x}_j'\mathbf{x}_j - \mathbf{x}_j'X_{-j}(X_{-j}'X_{-j})^{-1}X_{-j}'\mathbf{x}_j\}^{-1} = \{\mathbf{x}_j'(I - H_{-j})\mathbf{x}_j\}^{-1},$$

the inverse residual sum of squares of $\mathbf{x}_j$ on X_{-j}, which is $1 - R_j^2$ since $\mathbf{x}_j'\mathbf{x}_j = 1$. ∎

$\mathrm{VIF}_j = (1-R_j^2)^{-1}$ is the **variance inflation factor**: the factor by which $\operatorname{Var}\hat\beta_j$ exceeds its value under orthogonal regressors. $\mathrm{VIF}_j > 10$ (i.e. $R_j^2 > 0.9$) is a common warning sign.

15.2 Ridge regression

Motivation. Ridge adds kI to $X^\top X$: biased, but for suitable k smaller MSE than least squares.

Measure the accuracy of an estimator b by the total mean squared error $E\|b-\beta\|^2$. For least squares,

$$E\|\hat\beta-\beta\|^2=\operatorname{tr}\operatorname{Var}\hat\beta=\sigma^2\sum_j\frac{1}{\lambda_j}.$$

Definition (Ridge estimator)

For $k>0$, $\hat\beta(k)=(X'X+kI_p)^{-1}X'\mathbf{y}$, the minimizer of $\|\mathbf{y}-X\beta\|^2+k\|\beta\|^2$.

Adding k lifts all eigenvalues, conditioning $X'X+kI$. If $X'X=I$, $\hat\beta(k)=\hat\beta/(1+k)$: ridge **shrinks** towards **0**. In general $\hat\beta(k)=(X'X+kI)^{-1}X'X\hat\beta$, with $E\hat\beta(k)=(X'X+kI)^{-1}X'X\beta\neq\beta$.

Canonical form. Let $X'X=P\Lambda P'$ with P orthogonal and $\Lambda=\operatorname{diag}(\lambda_1,\dots,\lambda_p)$. With $Z=XP$ and $\alpha=P'\beta$, the model is $\mathbf{y}=Z\alpha+\epsilon$ with $Z'Z=\Lambda$. Then $\hat\alpha=\Lambda^{-1}Z'\mathbf{y}$, $\operatorname{Var}\hat\alpha=\sigma^2\Lambda^{-1}$, and

$$\hat\alpha(k)=P'\hat\beta(k)=(\Lambda+kI)^{-1}Z'\mathbf{y},\qquad \hat\alpha_j(k)=\frac{\lambda_j}{\lambda_j+k}\hat\alpha_j:$$

each canonical coefficient is shrunk, most strongly in directions with small λ_j. Since P is orthogonal, $\|\hat\beta(k)-\beta\|=\|\hat\alpha(k)-\alpha\|$.

Theorem 15.2 (Mean squared error of ridge)

$$E\|\hat\beta(k)-\beta\|^2=\sigma^2\sum_{j=1}^p\frac{\lambda_j}{(\lambda_j+k)^2}+k^2\sum_{j=1}^p\frac{\alpha_j^2}{(\lambda_j+k)^2}.$$

The first term (total variance) decreases in k and the second (squared bias) increases. There is always a $k>0$ with $E\|\hat\beta(k)-\beta\|^2<E\|\hat\beta-\beta\|^2$.

Proof [Supplement]. $\hat\alpha_j(k)$ has mean $\frac{\lambda_j}{\lambda_j+k}\alpha_j$ and variance $\frac{\lambda_j^2}{(\lambda_j+k)^2}\cdot\frac{\sigma^2}{\lambda_j}$; its bias is $-\frac{k}{\lambda_j+k}\alpha_j$. Adding variance and squared bias over j gives the formula. Call it $g(k)$. Then $g(0)=\sigma^2\sum 1/\lambda_j=E\|\hat\beta-\beta\|^2$, and

$$g'(k)=-2\sigma^2\sum_j\frac{\lambda_j}{(\lambda_j+k)^3}+2k\sum_j\frac{\lambda_j\alpha_j^2}{(\lambda_j+k)^3}.$$

At $k=0$, $g'(0)=-2\sigma^2\sum_j\lambda_j^{-2}<0$, so $g(k)<g(0)$ for small $k>0$. ■

In the orthonormal case $X'X=I$, $g(k)=\frac{p\sigma^2+k^2\|\alpha\|^2}{(1+k)^2}$, minimized at $k=p\sigma^2/\|\beta\|^2$ (set $g'(k)=0$). More generally, a separate constant k_j for each canonical direction, $\hat\alpha_j=\frac{\lambda_j}{\lambda_j+k_j}\hat\alpha_j$ (generalized ridge), has mean squared error $\sum_j\frac{\sigma^2\lambda_j+k_j^2\alpha_j^2}{(\lambda_j+k_j)^2}$, minimized at $k_j=\sigma^2/\alpha_j^2$.

Choosing k. The optimal k depends on unknown σ^2 and β. Practical choices:

- the **ridge trace**: plot $\hat\beta_j(k)$ against k and take the smallest k at which the coefficients stabilize;
- the plug-in rule $\hat k=p\hat\sigma^2/\hat\beta'\hat\beta$ (Hoerl, Kennard and Baldwin), possibly iterated with $\hat\beta(k)$ in place of $\hat\beta$;
- cross-validation, e.g. minimizing $\text{PRESS}(k)=\sum_i\{y_i-\mathbf{x}_i'\hat\beta_{(i)}(k)\}^2$. As in Theorem 10.3, with $A(k)=X(X'X+kI)^{-1}X'=(a_{ij})$ and $e_i=y_i-\mathbf{x}_i'\hat\beta(k)$,

$$\text{PRESS}(k)=\sum_{i=1}^n\Big(\frac{e_i}{1-a_{ii}}\Big)^2,$$

so no refitting is needed (leave-one-out cross-validation).

15.3 Principal component regression

Let $\mathbf{e}_1, \ldots, \mathbf{e}_p$ be orthonormal eigenvectors of $X'X$ with eigenvalues $l_1 \geq \cdots \geq l_p > 0$, and $E_r = (\mathbf{e}_1, \ldots, \mathbf{e}_r)$. The **principal components** are the columns of $Z_r = XE_r$, and they are orthogonal: $Z_r'Z_r = \operatorname{diag}(l_1, \ldots, l_r)$.

All components ($r = p$). $X\beta = XE_pE_p'\beta = Z\alpha$ with $\alpha = E_p'\beta$, and

$$\hat{\beta} = E_p\hat{\alpha} = E_p(Z'Z)^{-1}Z'\mathbf{y} = \sum_{k=1}^{p} \frac{1}{l_k}\mathbf{e}_k\mathbf{e}_k'X'\mathbf{y}, \qquad \operatorname{Var}\hat{\beta} = \sigma^2 \sum_{k=1}^{p} \frac{1}{l_k}\mathbf{e}_k\mathbf{e}_k',$$

which is just least squares.

Leading components ($r < p$). Regress $\mathbf{y}$ on Z_r only and transform back:

$$\tilde{\beta}_r = E_r(Z_r'Z_r)^{-1}Z_r'\mathbf{y} = \sum_{k=1}^{r} \frac{1}{l_k}\mathbf{e}_k\mathbf{e}_k'X'\mathbf{y}.$$

Theorem 15.3 (Principal component regression)

$$E\tilde{\beta}_r = \beta - \sum_{k=r+1}^{p} \mathbf{e}_k\mathbf{e}_k'\beta, \qquad \operatorname{Var}\tilde{\beta}_r = \sigma^2 \sum_{k=1}^{r} \frac{1}{l_k}\mathbf{e}_k\mathbf{e}_k',$$

and

$$E\|\tilde{\beta}_r - \beta\|^2 = \sigma^2 \sum_{k=1}^{r} \frac{1}{l_k} + \sum_{k=r+1}^{p} (\mathbf{e}_k'\beta)^2.$$

Proof. Using $X'X = \sum_h l_h\mathbf{e}_h\mathbf{e}_h'$ and orthonormality, $E\tilde{\beta}_r = \sum_{k\leq r} \frac{1}{l_k}\mathbf{e}_k\mathbf{e}_k'X'X\beta = \sum_{k\leq r} \mathbf{e}_k\mathbf{e}_k'\beta$, and $\beta = \sum_{k\leq p} \mathbf{e}_k\mathbf{e}_k'\beta$. Next,

$$\operatorname{Var}\tilde{\beta}_r = \sigma^2 \sum_{k,k'\leq r} \frac{1}{l_k l_{k'}}\mathbf{e}_k\mathbf{e}_k'X'X\mathbf{e}_{k'}\mathbf{e}_{k'}' = \sigma^2 \sum_{k\leq r} \frac{1}{l_k}\mathbf{e}_k\mathbf{e}_k'.$$

The trace of the variance plus the squared length of the bias gives the mean squared error. ■

Dropping the smallest-l_k components removes the largest variances σ^2/l_k at bias $\sum_{k>r}(\mathbf{e}_k'\beta)^2$, worthwhile when those directions carry little of β. PCR is "hard" ridge: ridge shrinks each canonical coefficient by $l_k/(l_k + k)$, PCR keeps r and zeros the rest.

Choosing r. Keep components until $\sum_{k\leq r} l_k / \sum_k l_k$ is large, look for a scree-plot elbow, or cross-validate. Components ignore y; partial least squares builds components correlated with y.

15.4 Selected exercises

Motivation. The VIF on the original scale, ridge as least squares on augmented data, and ridge as norm-constrained least squares.

Exercise 15.1 (The VIF on the original scale)

In $Y_i = \beta_0+\beta_1 x_{i1}+\cdots+\beta_p x_{ip}+\epsilon_i$ (uncorrelated errors, variance σ^2, unstandardized regressors), let $S_{jj} = \sum_i (x_{ij} - \bar{x}_j)^2 > 0$ and R_j^2 the R^2 of x_j on the other regressors (with intercept). Show that

$$\operatorname{Var}\hat{\beta}_j = \frac{\sigma^2}{S_{jj}}\,\mathrm{VIF}_j, \qquad \mathrm{VIF}_j = (1-R_j^2)^{-1}.$$

Solution.

Step 1. (Centring.) Let $C = I - J_n/n$, the projection onto $\mathbf{1}^\perp$, and $\tilde{X} = CX_s$, where X_s holds the p regressor columns. By Theorem 6.5 with $X_1 = \mathbf{1}$, the slope vector is $\hat{\beta}_s = (\tilde{X}'\tilde{X})^{-1}\tilde{X}'C\mathbf{y} = (\tilde{X}'\tilde{X})^{-1}\tilde{X}'\mathbf{y}$, so $\operatorname{Var}\hat{\beta}_s = (\tilde{X}'\tilde{X})^{-1}\tilde{X}'(\sigma^2 I)\tilde{X}(\tilde{X}'\tilde{X})^{-1} = \sigma^2(\tilde{X}'\tilde{X})^{-1}$.

Step 2. (Partitioned inverse.) Write $\tilde{X} = (\tilde{\mathbf{x}}_j, \tilde{X}_{-j})$ and $\tilde{H}_{-j} = \tilde{X}_{-j}(\tilde{X}'_{-j}\tilde{X}_{-j})^{-1}\tilde{X}'_{-j}$. As in the proof of Theorem 15.1, the j-th diagonal entry of $(\tilde{X}'\tilde{X})^{-1}$ is $\{\tilde{\mathbf{x}}_j'(I-\tilde{H}_{-j})\tilde{\mathbf{x}}_j\}^{-1}$.

Step 3. (A residual sum of squares.) $(\mathbf{1}, X_{-j})$ has projection $J_n/n + \tilde{H}_{-j}$ (centred $\tilde{X}_{-j} \perp \mathbf{1}$; cf. Theorem 10.1(d)), and $\tilde{H}_{-j}\mathbf{1} = \mathbf{0}$, so the residual of $\mathbf{x}_j$ is

$$(I - J_n/n - \tilde{H}_{-j})\mathbf{x}_j = \tilde{\mathbf{x}}_j - \tilde{H}_{-j}\tilde{\mathbf{x}}_j = (I-\tilde{H}_{-j})\tilde{\mathbf{x}}_j,$$

whose squared length is $\tilde{\mathbf{x}}_j'(I-\tilde{H}_{-j})\tilde{\mathbf{x}}_j$. The total sum of squares of x_j is S_{jj}, so this residual sum of squares equals $S_{jj}(1-R_j^2)$.

Step 4. So $\operatorname{Var}\hat{\beta}_j = \sigma^2/\{S_{jj}(1-R_j^2)\}$: σ^2/S_{jj} is the simple-regression variance (Theorem 4.1(a)), attained when x_j is uncorrelated with the others, and VIF_j the collinearity inflation; $S_{jj} = 1$ gives Theorem 15.1. ∎

Exercise 15.2 (Ridge regression as augmented least squares)

Let $E(\mathbf{y} \mid X) = X\beta$ and $\operatorname{Var}(\mathbf{y} \mid X) = \sigma^2 I_n$, X of size $n \times p$ with singular value decomposition $X = UDV'$ (U of size $n \times p$ with orthonormal columns, $D = \operatorname{diag}(d_1, \ldots, d_p)$ with $d_j \geq 0$, V orthogonal). For $\lambda > 0$ let

$$X_* = \begin{pmatrix} X \\ \sqrt{\lambda} I_p \end{pmatrix}, \qquad \mathbf{y}_* = \begin{pmatrix} \mathbf{y} \\ \mathbf{0}_p \end{pmatrix}, \qquad \hat{\beta} = (X_*'X_*)^{-1}X_*'\mathbf{y}_*.$$

Express $E(\hat{\beta} \mid X)$ and $\operatorname{Var}(\hat{\beta} \mid X)$ through U, D, V, β and σ.

Solution.

Step 1. $X_*'X_* = X'X + \lambda I$ and $X_*'\mathbf{y}_* = X'\mathbf{y}$, so $\hat{\beta} = (X'X + \lambda I)^{-1}X'\mathbf{y}$ is the ridge estimator with $k = \lambda$; it is defined even if X is rank deficient, because $X'X + \lambda I$ is positive definite.

Step 2. $X'X = VDU'UDV' = VD^2V'$ and $X'X + \lambda I = V(D^2 + \lambda I)V'$, so $(X'X + \lambda I)^{-1} = V(D^2 + \lambda I)^{-1}V'$ and

$$\hat{\beta} = V(D^2 + \lambda I)^{-1}V'VDU'\mathbf{y} = V(D^2 + \lambda I)^{-1}DU'\mathbf{y}.$$

Step 3. (Mean.) $E(\hat{\beta} \mid X) = V(D^2 + \lambda I)^{-1}DU'UDV'\beta = V\operatorname{diag}\Big(\frac{d_j^2}{d_j^2 + \lambda}\Big)V'\beta.$

Step 4. (Variance.) With $M = V(D^2 + \lambda I)^{-1}DU'$, $\operatorname{Var}(\hat{\beta} \mid X) = \sigma^2 MM'$ and $U'U = I$ give

$$\operatorname{Var}(\hat{\beta} \mid X) = \sigma^2 V(D^2 + \lambda I)^{-1}D^2(D^2 + \lambda I)^{-1}V' = \sigma^2 V\operatorname{diag}\Big(\frac{d_j^2}{(d_j^2 + \lambda)^2}\Big)V'.$$

In coordinates $\alpha = V'\beta$ these are the canonical formulas (Section 15.2) with $\lambda_j = d_j^2$: shrinkage by $d_j^2/(d_j^2+\lambda)$. The p extra rows are pseudo-observations $0 = \sqrt{\lambda}\beta_j + \text{error}$ pulling β towards $\mathbf{0}$. ■

Exercise 15.3 (Ridge as constrained least squares)

Let X $(n\times p)$ have full column rank, $\hat\beta$ be the least squares estimator, $Q(\beta) = (\beta-\hat\beta)'X'X(\beta-\hat\beta)$, and $c > 0$.
(a) Find the minimizer of $Q(\beta)$ subject to $\beta'\beta \le c$.
(b) For fixed $k \ge 0$, find the mean and variance of $\hat\beta(k) = (X'X+kI)^{-1}X'\mathbf{y}$.
(c) Using the singular value decomposition $X = UDV'$, show that $\hat\beta(k)$ shrinks $\hat\beta$ towards $\mathbf{0}$.

Solution.

(a) If $\hat\beta'\hat\beta \le c$, then $\hat\beta$ is feasible with $Q(\hat\beta) = 0$, and it is the unique minimizer since $X'X$ is positive definite. Assume now $\hat\beta'\hat\beta > c$.

Step 1. For $k \ge 0$, $Q(\beta)+k\|\beta\|^2$ is strictly convex with gradient $2X'X(\beta-\hat\beta)+2k\beta$, which vanishes at $(X'X+kI)^{-1}X'X\hat\beta = (X'X+kI)^{-1}X'\mathbf{y} = \hat\beta(k)$. So $\hat\beta(k)$ is its unique minimizer.

Step 2. With $X'X = P\Lambda P'$ and $\hat\alpha = P'\hat\beta$, $\|\hat\beta(k)\|^2 = \sum_j \lambda_j^2\hat\alpha_j^2/(\lambda_j+k)^2$. This is continuous and strictly decreasing in k (since $\hat\alpha \ne \mathbf{0}$), equals $\|\hat\beta\|^2 > c$ at $k = 0$, and tends to 0. Hence there is a unique $k^* > 0$ with $\|\hat\beta(k^*)\|^2 = c$.

Step 3. For any feasible $\beta \ne \hat\beta(k^*)$, by Step 1,

$$Q(\beta) > Q(\hat\beta(k^*)) + k^*(\|\hat\beta(k^*)\|^2 - \|\beta\|^2) = Q(\hat\beta(k^*)) + k^*(c - \|\beta\|^2) \ge Q(\hat\beta(k^*)).$$

So the constrained minimizer is the ridge estimator $\hat\beta(k^*)$, the ridge constant being the Lagrange multiplier of the constraint.

(b) $\hat\beta(k) = M\mathbf{y}$ with $M = (X'X+kI)^{-1}X'$, so

$$E\hat\beta(k) = (X'X+kI)^{-1}X'X\beta, \qquad \operatorname{Var}\hat\beta(k) = \sigma^2(X'X+kI)^{-1}X'X(X'X+kI)^{-1}.$$

These formulas are for a fixed k; the k^* of (a) depends on $\mathbf{y}$, so they describe the constrained estimator only conditionally on its multiplier.

(c) As in Exercise 15.2, $\hat\beta(k) = V\,\mathrm{diag}\{d_j^2/(d_j^2+k)\}V'\hat\beta$, $d_j > 0$; each coordinate of $V'\hat\beta$ is multiplied by a factor in $(0,1)$ for $k > 0$, so $\|\hat\beta(k)\| < \|\hat\beta\|$ $(\hat\beta \ne \mathbf{0})$, with the smallest singular values shrunk most. ■